

Problem B (Bulgarian solitaire), Cell C1: Cyclic Partitions and Cycles

Written proof

Submission of team record B-C1

Cell status: SOLVED **Claim status:** PROVED

Established: parts (i), (ii)(a) and (ii)(b) of Cell C1, for every $k \geq 1$ and every $1 \leq r \leq k$. The proof is purely written; the included script (Section 6) is a sanity check for $1 \leq n \leq 45$ and the proof does not rely on it.

1 Statement

Definitions (verbatim from the official problem statement)

A *partition* of a positive integer n is a weakly decreasing sequence

$$\lambda = (\lambda_1, \dots, \lambda_s)$$

of positive integers with

$$\lambda_1 + \dots + \lambda_s = n.$$

Think of it as a division of n cards into s piles.

The *shift* $B(\lambda)$ is the partition of n obtained as follows: remove one card from every pile, discard the piles that become empty, and add one new pile consisting of the s removed cards. Formally, $B(\lambda)$ is the partition whose parts are the positive numbers among

$$\lambda_1 - 1, \dots, \lambda_s - 1$$

together with one extra part equal to s .

Call λ *cyclic* if

$$B^i(\lambda) = \lambda \quad \text{for some } i \geq 1.$$

Write

$$T_k = \frac{k(k+1)}{2}, \quad \delta_k = (k, k-1, \dots, 2, 1),$$

the partition of T_k into distinct parts. Every $n \geq 1$ satisfies

$$T_{k-1} < n \leq T_k$$

for exactly one k , which we call the *rank* of n .

Cell C1 (verbatim)

First, the long-run behaviour: which partitions repeat under the shift, and how they fall into cycles. Let $n = T_k$. Prove that for every partition λ of n there is an i with $B^i(\lambda) = \delta_k$, and that δ_k is the only cyclic partition of n . Then let n be arbitrary of rank k , say $n = T_{k-1} + r$ with $1 \leq r \leq k$: determine all cyclic partitions of n , and determine the number of distinct cycles of B on the partitions of n . Prove both.

Exact targets (as fixed in the team's target file)

- (i) For every $k \geq 1$ and $n = T_k$: for every partition λ of n there is an $i \geq 0$ with $B^i(\lambda) = \delta_k$; and δ_k is the only cyclic partition of n .
- (ii) For every $k \geq 1$ and every r with $1 \leq r \leq k$, $n = T_{k-1} + r$:
 - (a) give an explicit description (as functions of k and r) of the set of *all* cyclic partitions of n , and prove that a partition of n is cyclic if and only if it lies in that set;
 - (b) give the exact number of distinct cycles of B on the partitions of n (as a formula in k and r), and prove it.

2 Status

- **Cell status:** SOLVED.
- **Claim status:** PROVED.

3 Answer

Let $k \geq 1$, $1 \leq r \leq k$, $n = T_{k-1} + r$.

- (ii)(a) A partition of n is cyclic if and only if it equals

$$\lambda(\varepsilon) := (k-1+\varepsilon_1, k-2+\varepsilon_2, \dots, 1+\varepsilon_{k-1}, \varepsilon_k) \quad (\text{last entry deleted when it is } 0)$$

for some $\varepsilon \in \{0, 1\}^k$ with $\varepsilon_1 + \dots + \varepsilon_k = r$. Equivalently: the Young diagram is the staircase δ_{k-1} plus exactly r of the k cells of the k -th diagonal. There are exactly $\binom{k}{r}$ cyclic partitions of n , and B acts on them by $B(\lambda(\varepsilon)) = \lambda(\varepsilon_k, \varepsilon_1, \dots, \varepsilon_{k-1})$.

- (ii)(b) The number of distinct cycles of B on the partitions of n is

$$N(k, r) = \frac{1}{k} \sum_{d \mid \gcd(k, r)} \varphi(d) \binom{k/d}{r/d}.$$

- (i) (Case $r = k$, $n = T_k$.) δ_k is the only cyclic partition of T_k , and every partition λ of T_k has $B^i(\lambda) = \delta_k$ for some $i \geq 0$.

4 Cited results vs. our contribution

Cited (standard results used by name in the proof).

- Pigeonhole principle (step 1.2).
- Chinese remainder theorem for the coprime moduli $d, d + 1$ (proof of Lemma 6).
- Orbit-counting lemma (Burnside’s lemma), step 10.2; the proof includes a short derivation of it from the orbit–stabiliser theorem, which is used without proof.
- Euler’s totient function φ (convention 0.1); the identity $\sum_{d|k} \varphi(d) = k$ used in the check after 10.4 is justified there by partitioning $\{1, \dots, k\}$ by $\gcd(\cdot, k)$.

The accepted artefacts give no bibliographic references for these standard facts, and none are added here. The accepted artefacts cite no literature on Bulgarian solitaire, and no literature search is part of the material this document was prepared from; this document therefore makes no claim either way about whether these results appear in the literature.

Our contribution (the team’s accepted artefacts).

- The written proof below (Sections 0–11 of the proof), establishing claims R1–R11, all with claim status PROVED.
- The sanity-check script `code/check_c1.py` (stdlib only, exact integer arithmetic), which confirms (i), (ii)(a), (ii)(b) by brute force for all $1 \leq n \leq 45$. It is not load-bearing.

5 Proof

The text of this section is the team’s accepted proof, typeset from Markdown to L^AT_EX with its numbering preserved. The only editorial change is the script path in the final paragraph (see Section 7).

Statement proved. For every $k \geq 1$ and every $1 \leq r \leq k$, with $n = T_{k-1} + r$: a partition of n is cyclic if and only if it equals $\lambda(\varepsilon) := (k-1+\varepsilon_1, k-2+\varepsilon_2, \dots, 1+\varepsilon_{k-1}, \varepsilon_k)$ (last entry deleted when it is 0) for some $\varepsilon \in \{0, 1\}^k$ with $\varepsilon_1 + \dots + \varepsilon_k = r$; B acts on these by $B(\lambda(\varepsilon)) = \lambda(\varepsilon_k, \varepsilon_1, \dots, \varepsilon_{k-1})$; the number of distinct cycles of B on the partitions of n is $N(k, r) = \frac{1}{k} \sum_{d|\gcd(k, r)} \varphi(d) \binom{k/d}{r/d}$. In particular (case $r = k$, $n = T_k$) δ_k is the only cyclic partition of T_k and every partition λ of T_k has $B^i(\lambda) = \delta_k$ for some $i \geq 0$.

0. Conventions

0.1. Reading of “cycle of B ”: a *cycle* is a set $\{\mu, B(\mu), \dots, B^{p-1}(\mu)\}$ where $B^p(\mu) = \mu$, $p \geq 1$ (the forward orbit of a cyclic partition). Two cycles are distinct if they are distinct as sets. φ is Euler’s totient function. $\mathbb{N}_{\geq 1} = \{1, 2, 3, \dots\}$.

0.2. For a finite sequence $c = (c_1, \dots, c_m)$ of positive integers (not necessarily ordered) put

$$\mathcal{C}(c) = \{(j, h) \in \mathbb{N}_{\geq 1}^2 : 1 \leq j \leq m, 1 \leq h \leq c_j\}$$

(pile j , height h), so $|\mathcal{C}(c)| = c_1 + \dots + c_m$. A partition is determined by its cell set: $\lambda_j = |\{h : (j, h) \in \mathcal{C}(\lambda)\}|$.

0.3. For a partition ν : if $(1, h) \in \mathcal{C}(\nu)$ and $1 \leq h' \leq h$ then $(1, h') \in \mathcal{C}(\nu)$ (definition of $\mathcal{C}$ with $j = 1$).

0.4. Diagonals: for $d \geq 1$ let $D_d = \{(j, h) \in \mathbb{N}_{\geq 1}^2 : j + h - 1 = d\} = \{(j, d + 1 - j) : 1 \leq j \leq d\}$, so $|D_d| = d$, and every cell of $\mathbb{N}_{\geq 1}^2$ lies in exactly one D_d (namely $d = j + h - 1 \geq 1$).

0.5. Weight: $E(c) = \sum_{(j,h) \in \mathcal{C}(c)} (j + h - 1)$. Summing over h first,

$$E(c) = \sum_{j=1}^m \sum_{h=1}^{c_j} (j - 1 + h) = \sum_{j=1}^m (j - 1) c_j + \sum_{j=1}^m \frac{c_j(c_j + 1)}{2}.$$

0.6. Unsorted shift: for a partition $\lambda = (\lambda_1, \dots, \lambda_s)$ of n let $t = |\{j : \lambda_j \geq 2\}|$. Because λ is weakly decreasing, $\lambda_j \geq 2 \iff j \leq t$. Put

$$U(\lambda) = (s, \lambda_1 - 1, \dots, \lambda_t - 1),$$

a sequence of $t + 1$ positive integers. Its entries are exactly the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with s , so by the definition of the shift, $B(\lambda)$ is the weakly decreasing rearrangement of $U(\lambda)$; write $B(\lambda) = \text{sort}(U(\lambda))$.

0.7. The map $R : \mathbb{N}_{\geq 1}^2 \rightarrow \mathbb{N}_{\geq 1}^2$: $R(j, h) = (j + 1, h - 1)$ if $h \geq 2$, and $R(j, 1) = (1, j)$.

1. B is a self-map; orbits reach cyclic partitions (R1)

1.1. $B(\lambda)$ is a partition of n : its parts are positive (the $\lambda_j - 1$ kept are positive and $s \geq 1$), it is weakly decreasing by 0.6, and its sum is $\sum_{j \leq t} (\lambda_j - 1) + s = \sum_{j \leq s} (\lambda_j - 1) + s = n$, since $\lambda_j - 1 = 0$ for $j > t$.

1.2. The set $P(n)$ of partitions of n is finite (each is a sequence of at most n entries from $\{1, \dots, n\}$). For $\lambda \in P(n)$, the elements $B^0(\lambda), B^1(\lambda), \dots, B^{|P(n)|}(\lambda)$ are $|P(n)| + 1$ elements of $P(n)$, so by the pigeonhole principle $B^a(\lambda) = B^b(\lambda)$ for some $0 \leq a < b$. Then $\mu = B^a(\lambda)$ satisfies $B^{b-a}(\mu) = \mu$ with $b - a \geq 1$, i.e. μ is cyclic.

2. The unsorted shift moves cells by R (R2)

2.1. Claim: R restricts to a bijection $\mathcal{C}(\lambda) \rightarrow \mathcal{C}(U(\lambda))$. Write $U(\lambda) = (c_1, \dots, c_{t+1})$ with $c_1 = s$, $c_{j+1} = \lambda_j - 1$ ($1 \leq j \leq t$).

- If $(j, h) \in \mathcal{C}(\lambda)$ with $h \geq 2$: then $\lambda_j \geq h \geq 2$, so $j \leq t$ (0.6), and $R(j, h) = (j + 1, h - 1)$ has $2 \leq j + 1 \leq t + 1$ and $1 \leq h - 1 \leq \lambda_j - 1 = c_{j+1}$; so $R(j, h) \in \mathcal{C}(U(\lambda))$ with first coordinate ≥ 2 .
- If $(j, 1) \in \mathcal{C}(\lambda)$: $1 \leq j \leq s = c_1$, so $R(j, 1) = (1, j) \in \mathcal{C}(U(\lambda))$ with first coordinate 1.
- Inverse map $\mathcal{C}(U(\lambda)) \rightarrow \mathcal{C}(\lambda)$: $(1, h) \mapsto (h, 1)$ (valid: $h \leq c_1 = s$, and $\lambda_h \geq 1$); $(j', h') \mapsto (j' - 1, h' + 1)$ for $j' \geq 2$ (valid: $j' - 1 \leq t$ and $h' + 1 \leq c_{j'} + 1 = \lambda_{j'-1}$). Composing in both orders gives the identity (check on the two cases $h = 1$ / $h \geq 2$, respectively $j' = 1$ / $j' \geq 2$). So $R : \mathcal{C}(\lambda) \rightarrow \mathcal{C}(U(\lambda))$ is a bijection.

2.2. R preserves $j + h - 1$: $(j + 1) + (h - 1) - 1 = j + h - 1$ and $1 + j - 1 = j + 1 - 1$. Hence $R(D_d) \subseteq D_d$ for each d , and by 2.1, $E(U(\lambda)) = E(\lambda)$.

3. Sorting lemma (R3)

Lemma 3. *Let $c = (c_1, \dots, c_m)$ be positive integers with sum n and let $\lambda = \text{sort}(c)$ be its weakly decreasing rearrangement. Then $E(\lambda) \leq E(c)$, with equality if and only if c is weakly decreasing (i.e. $c = \lambda$).*

3.1. By 0.5 and since $\sum_j c_j(c_j + 1)/2$ is unchanged by rearranging, $E(c) - E(\lambda) = \sum_j (j - 1)c_j - \sum_j (j - 1)\lambda_j$.

3.2. Exchanging the order of summation, $\sum_{j=1}^m (j - 1)c_j = \sum_{j=1}^m \sum_{i=1}^{j-1} c_j = \sum_{i=1}^{m-1} \sum_{j=i+1}^m c_j = \sum_{i=1}^{m-1} (n - P_i(c))$ with $P_i(c) = c_1 + \dots + c_i$. The same identity holds for λ .

3.3. For each i , $P_i(c) \leq P_i(\lambda)$: there is a permutation σ with $c_l = \lambda_{\sigma(l)}$; list $\sigma(1), \dots, \sigma(i)$ increasingly as $j_1 < \dots < j_i$; then $j_l \geq l$, so $\lambda_{j_l} \leq \lambda_l$ (λ weakly decreasing), and summing, $P_i(c) = \sum_l \lambda_{j_l} \leq \sum_{l \leq i} \lambda_l = P_i(\lambda)$.

3.4. By 3.2 and 3.3, $E(c) - E(\lambda) = \sum_{i=1}^{m-1} (P_i(\lambda) - P_i(c)) \geq 0$.

3.5. Equality case. If c is weakly decreasing then $c = \lambda$ (the weakly decreasing rearrangement of a multiset is unique) and equality holds. If c is not weakly decreasing, pick i with $c_i < c_{i+1}$ ($1 \leq i \leq m - 1$). The i entries $c_1, \dots, c_{i-1}, c_{i+1}$ (at distinct positions) have sum $P_i(c) - c_i + c_{i+1} > P_i(c)$, and by the argument of 3.3 applied to these i positions, that sum is $\leq P_i(\lambda)$. So $P_i(c) < P_i(\lambda)$, and by 3.4 $E(c) > E(\lambda)$. $\square$

4. Energy along orbits; no sorting on cycles (R4)

4.1. For every partition λ : $E(B(\lambda)) = E(\text{sort}(U(\lambda))) \leq E(U(\lambda)) = E(\lambda)$ (Lemma 3, then 2.2), with equality iff $U(\lambda)$ is weakly decreasing, in which case $B(\lambda) = U(\lambda)$ as sequences and so, by 2.1, $\mathcal{C}(B(\lambda)) = R(\mathcal{C}(\lambda))$.

4.2. Let λ be cyclic, $B^p(\lambda) = \lambda$, $p \geq 1$. By 4.1, $E(\lambda) \geq E(B\lambda) \geq \dots \geq E(B^p\lambda) = E(\lambda)$, so $E(B^{i+1}\lambda) = E(B^i\lambda)$ for $0 \leq i < p$. For arbitrary $i \geq 0$, $B^i\lambda = B^{i \bmod p}\lambda$ (induction on i using $B^p\lambda = \lambda$), so $E(B^{i+1}\lambda) = E(B^i\lambda)$ for all $i \geq 0$. By the equality case of 4.1, $\mathcal{C}(B^{i+1}\lambda) = R(\mathcal{C}(B^i\lambda))$ for all $i \geq 0$, hence by induction

$$\mathcal{C}(B^t\lambda) = R^t(\mathcal{C}(\lambda)) \quad (t \geq 0).$$

5. R rotates each diagonal (R5)

5.1. Index D_d by its first coordinate: $x_a = (a, d + 1 - a)$, $1 \leq a \leq d$. For $a < d$ the height $d + 1 - a \geq 2$, so $R(x_a) = (a + 1, d - a) = x_{a+1}$; for $a = d$, $R(x_d) = R(d, 1) = (1, d) = x_1$. So $R|_{D_d}$ is the cyclic rotation $x_a \mapsto x_{a+1}$ (indices mod d in $\{1, \dots, d\}$), a bijection of D_d , and by induction on t , $R^t(x_a) = x_{a'}$ with $a' \equiv a + t \pmod{d}$.

5.2. Consequently, for cyclic λ and $t \geq 0$, by 4.2 and 5.1 (each R^t maps D_d bijectively onto itself and maps cells in different diagonals to different diagonals):

$$\mathcal{C}(B^t\lambda) \cap D_d = R^t(\mathcal{C}(\lambda) \cap D_d),$$

and for $x \in D_d$: $x \in \mathcal{C}(\lambda) \iff R^t(x) \in \mathcal{C}(B^t\lambda)$ (injectivity of R^t on D_d).

6. Key lemma (R6)

Lemma 6. *Let λ be cyclic and $d \geq 1$. If $D_d \not\subseteq \mathcal{C}(\lambda)$ then $D_{d+1} \cap \mathcal{C}(\lambda) = \emptyset$.*

Proof. Suppose $x_a = (a, d+1-a) \in D_d \setminus \mathcal{C}(\lambda)$ and $y_b = (b, d+2-b) \in D_{d+1} \cap \mathcal{C}(\lambda)$. Since $\gcd(d, d+1) = 1$, the Chinese remainder theorem gives $t \geq 0$ with $t \equiv 1-a \pmod{d}$ and $t \equiv 1-b \pmod{d+1}$. By 5.1, $R^t(x_a) = (1, d)$ and $R^t(y_b) = (1, d+1)$. By 5.2, $(1, d) \notin \mathcal{C}(B^t\lambda)$ and $(1, d+1) \in \mathcal{C}(B^t\lambda)$. But $B^t\lambda$ is a partition (1.1), and by 0.3, $(1, d+1) \in \mathcal{C}(B^t\lambda)$ forces $(1, d) \in \mathcal{C}(B^t\lambda)$. Contradiction. $\square$

7. Structure of cyclic partitions (R7)

7.1. Let λ be a cyclic partition of n . $\mathcal{C}(\lambda)$ is finite and D_d is nonempty, so $K = \min\{d \geq 1 : D_d \not\subseteq \mathcal{C}(\lambda)\}$ exists. $D_1 = \{(1, 1)\} \subseteq \mathcal{C}(\lambda)$ since $\lambda_1 \geq 1$, so $K \geq 2$.

7.2. Claim: $D_e \cap \mathcal{C}(\lambda) = \emptyset$ for all $e > K$. Induction on e . For $e = K+1$: Lemma 6 with $d = K$. For $e > K+1$: by induction $D_{e-1} \cap \mathcal{C}(\lambda) = \emptyset$; since $D_{e-1} \neq \emptyset$, $D_{e-1} \not\subseteq \mathcal{C}(\lambda)$, and Lemma 6 with $d = e-1$ gives $D_e \cap \mathcal{C}(\lambda) = \emptyset$.

7.3. By 0.4, 7.1, 7.2: $\mathcal{C}(\lambda) = D_1 \cup \dots \cup D_{K-1} \cup S$ with $S = \mathcal{C}(\lambda) \cap D_K \subsetneq D_K$, so $n = |\mathcal{C}(\lambda)| = T_{K-1} + |S|$ with $0 \leq |S| \leq K-1$.

7.4. Rank. $T_0 < T_1 < T_2 < \dots$ (as $T_m - T_{m-1} = m \geq 1$) and $T_0 = 0$, so the intervals $(T_{m-1}, T_m]$, $m \geq 1$, are disjoint and cover $\mathbb{N}_{\geq 1}$: the rank is well defined. Let $n = T_{k-1} + r$, $1 \leq r \leq k$, i.e. n has rank k .

- If $|S| \geq 1$: $T_{K-1} < n \leq T_{K-1} + K - 1 < T_K$, so the rank is K : $k = K$ and $r = n - T_{k-1} = |S|$.
- If $|S| = 0$: $n = T_{K-1}$ with $K-1 \geq 1$, and $T_{K-2} < n \leq T_{K-1}$, so $k = K-1$ and $r = n - T_{k-1} = T_k - T_{k-1} = k$; here $\mathcal{C}(\lambda) = D_1 \cup \dots \cup D_k$.

In both cases:

$$D_1 \cup \dots \cup D_{k-1} \subseteq \mathcal{C}(\lambda) \subseteq D_1 \cup \dots \cup D_k, \quad |\mathcal{C}(\lambda) \cap D_k| = r. \quad (*)$$

7.5. Reading off the parts. Define $\varepsilon_j = 1$ if $(j, k+1-j) \in \mathcal{C}(\lambda)$ and $\varepsilon_j = 0$ otherwise ($1 \leq j \leq k$); by (*), $\sum_j \varepsilon_j = r$. For $1 \leq j \leq k$ the cells (j, h) with $h \leq k-j$ lie in D_{j+h-1} with $j+h-1 \leq k-1$, so are in $\mathcal{C}(\lambda)$; the cell $(j, k+1-j) \in D_k$ is present iff $\varepsilon_j = 1$; cells (j, h) with $h \geq k+2-j$ lie in D_e , $e \geq k+1$, so are absent. Hence $\lambda_j = k-j + \varepsilon_j$ for $1 \leq j \leq k$ (where $\lambda_j := 0$ if $j > s$). For $j > k$, every cell (j, h) lies in D_e with $e \geq j \geq k+1$, so $\lambda_j = 0$. Therefore $\lambda = \lambda(\varepsilon)$ as defined in the statement line.

8. The converse and the action of B (R8)

8.1. Let $W(k, r) = \{\varepsilon \in \{0, 1\}^k : \sum_j \varepsilon_j = r\}$ and for $\varepsilon \in W(k, r)$ let $\lambda(\varepsilon)$ be the sequence of positive numbers among $k - j + \varepsilon_j$, $j = 1, \dots, k$, in this order. For $j \leq k - 1$, $k - j + \varepsilon_j \geq 1$, and the k -th value is ε_k , so only the last entry can be dropped (exactly when $\varepsilon_k = 0$). The sequence is weakly decreasing since $(k - j + \varepsilon_j) - (k - j - 1 + \varepsilon_{j+1}) = 1 + \varepsilon_j - \varepsilon_{j+1} \geq 0$, and its sum is $\sum_{j=1}^k (k - j) + r = T_{k-1} + r = n$. So $\lambda(\varepsilon)$ is a partition of n .

8.2. Injectivity: from $\lambda = \lambda(\varepsilon)$ (with $\lambda_j := 0$ beyond its length) we recover $\varepsilon_j = \lambda_j - (k - j)$, $1 \leq j \leq k$.

8.3. Let $\rho : W(k, r) \rightarrow W(k, r)$, $\rho(\varepsilon) = (\varepsilon_k, \varepsilon_1, \dots, \varepsilon_{k-1})$. Claim: $B(\lambda(\varepsilon)) = \lambda(\rho\varepsilon)$. Write $\lambda = \lambda(\varepsilon)$, with $s = k - 1 + \varepsilon_k$ parts. The parts of $B(\lambda)$ are $s = k - 1 + \varepsilon_k$ together with the positive values among $\lambda_j - 1$: for $1 \leq j \leq k - 1$, $\lambda_j - 1 = k - 1 - j + \varepsilon_j$; if $\varepsilon_k = 1$ there is also $\lambda_k - 1 = 0$, which is not positive. So the multiset of parts of $B(\lambda)$ is $\{k - 1 + \varepsilon_k\} \cup \{k - (j + 1) + \varepsilon_j : 1 \leq j \leq k - 1, \text{ value} > 0\}$. With $\varepsilon' = \rho\varepsilon$ ($\varepsilon'_1 = \varepsilon_k$, $\varepsilon'_{j+1} = \varepsilon_j$), the parts of $\lambda(\varepsilon')$ are the positive values among $k - 1 + \varepsilon'_1 = k - 1 + \varepsilon_k$ (positive, as $k - 1 + \varepsilon_k = s \geq 1$) and $k - (j + 1) + \varepsilon'_{j+1} = k - (j + 1) + \varepsilon_j$, $1 \leq j \leq k - 1$: the same multiset. A partition is determined by its multiset of parts, so $B(\lambda(\varepsilon)) = \lambda(\rho\varepsilon)$. (For $k = 1$: $W(1, 1) = \{(1)\}$, $\lambda = (1)$, $B((1)) = (1)$, consistent.)

8.4. By 8.3 and induction, $B^i(\lambda(\varepsilon)) = \lambda(\rho^i\varepsilon)$; since $\rho^k = \text{id}$ (each application moves every letter one place cyclically), $B^k(\lambda(\varepsilon)) = \lambda(\varepsilon)$ with $k \geq 1$, so $\lambda(\varepsilon)$ is cyclic.

9. Answer to (ii)(a) (R9)

Theorem 9. For $k \geq 1$, $1 \leq r \leq k$, $n = T_{k-1} + r$, the cyclic partitions of n are exactly the partitions $\lambda(\varepsilon)$, $\varepsilon \in W(k, r)$; equivalently, the partitions whose Young diagram consists of the staircase δ_{k-1} (all cells on diagonals $D_1, \dots, D_{k-1}$) plus exactly r of the k cells of diagonal D_k . There are exactly $\binom{k}{r}$ of them.

Proof. “Only if”: 7.5. “If”: 8.4. The count follows from the injectivity 8.2 and $|W(k, r)| = \binom{k}{r}$. $\square$

10. Answer to (ii)(b) (R10)

10.1. Let $\text{Cyc}(n)$ be the set of cyclic partitions and $\Lambda : W(k, r) \rightarrow \text{Cyc}(n)$, $\varepsilon \mapsto \lambda(\varepsilon)$: a bijection by Theorem 9 and 8.2, with $B \circ \Lambda = \Lambda \circ \rho$ (8.3). Every cycle (0.1) consists of cyclic partitions: if $B^p\mu = \mu$ then $B^p(B^i\mu) = B^i(B^p\mu) = B^i\mu$. For cyclic $\mu = \Lambda(\varepsilon)$, the cycle through μ is $\{B^i\mu : i \geq 0\} = \Lambda(\{\rho^i\varepsilon : 0 \leq i < k\})$, the image of the orbit of ε under the cyclic group $G = \langle \rho \rangle$ (of order dividing k ; we let G act through $j \mapsto \rho^j$, $j \in \mathbb{Z}/k$). Two cyclic partitions lie in the same cycle iff their ε 's lie in the same orbit, and orbits partition $W(k, r)$. Hence the number of distinct cycles equals the number of orbits of $\mathbb{Z}/k$ acting on $W(k, r)$ by $j \cdot \varepsilon = \rho^j\varepsilon$.

10.2. Orbit-counting lemma (Burnside's lemma, standard; proof included): for a finite group H acting on a finite set X , $\#\text{orbits} = \frac{1}{|H|} \sum_{g \in H} |\text{Fix}(g)|$. Proof: $\sum_g |\text{Fix}(g)| = |\{(g, x) : gx = x\}| = \sum_x |\text{Stab}(x)| = \sum_x |H|/|Hx|$ (orbit-stabiliser theorem), and $\sum_{x \in O} 1/|O| = 1$ for each orbit O .

10.3. Fixed points. Index positions by $\mathbb{Z}/k$; $(\rho^j \varepsilon)_i = \varepsilon_{i-j}$. So $\rho^j \varepsilon = \varepsilon$ iff $\varepsilon_i = \varepsilon_{i-j}$ for all i , iff ε is constant on each coset of the subgroup $\langle j \rangle \leq \mathbb{Z}/k$. With $g = \gcd(j, k)$ (and $\gcd(0, k) = k$), $\langle j \rangle = g\mathbb{Z}/k\mathbb{Z}$ has order k/g and there are g cosets. A word constant on cosets is determined by the set of cosets where it equals 1; it has r ones iff that set has $r/(k/g)$ elements. Hence $|\text{Fix}(\rho^j)| = \binom{g}{r_{g/k}}$ if $(k/g) \mid r$, and 0 otherwise.

10.4. For each divisor g of k , the $j \in \{0, \dots, k-1\}$ with $\gcd(j, k) = g$ are $j = gj'$ with $0 \leq j' < k/g$ and $\gcd(j', k/g) = 1$; there are $\varphi(k/g)$ of them. Substituting $d = k/g$ (so $d \mid k$ and the condition $(k/g) \mid r$ reads $d \mid r$):

$$\#\text{cycles} = N(k, r) = \frac{1}{k} \sum_{d \mid \gcd(k, r)} \varphi(d) \binom{k/d}{r/d}.$$

Checks: $r = k$ gives $\frac{1}{k} \sum_{d \mid k} \varphi(d) = 1$ (by $\sum_{d \mid k} \varphi(d) = k$: partition $\{1, \dots, k\}$ by $\gcd(\cdot, k)$ as in 10.4); $\gcd(k, r) = 1$ gives $\binom{k}{r}/k$. $\square$

10.5. (Remark, not needed.) The cycle through $\lambda(\varepsilon)$ has length equal to the least $p \geq 1$ with $\rho^p \varepsilon = \varepsilon$ (by 8.2–8.3), a divisor of k .

11. Part (i) (R11)

Let $n = T_k$, i.e. $r = k$ in Theorem 9 (rank of T_k is k by 7.4). $W(k, k) = \{(1, \dots, 1)\}$ and $\lambda(1, \dots, 1) = (k, k-1, \dots, 1) = \delta_k$, so δ_k is the only cyclic partition of T_k (and $B(\delta_k) = \delta_k$ by 8.3). For any partition λ of T_k , 1.2 gives $a \geq 0$ with $B^a(\lambda)$ cyclic, hence $B^a(\lambda) = \delta_k$. $\square$

Established / not established

Established: (i) and (ii)(a),(b) for all $k \geq 1$, $1 \leq r \leq k$, with the answer above. The brute-force script `code/check_c1.py` (all $n \leq 45$) agrees but the proof does not rely on it. Not addressed: time-to-cycle questions (cells C2 and beyond). Known gaps: none found.

6 How to verify

The proof. The proof above is self-contained and is to be checked by reading; it does not rely on any computation.

Sanity check (not load-bearing). The script `code/check_c1.py` (listed in full in the appendix; Python standard library only, exact integer arithmetic) does the following for every n in $[1, \text{NMAX}]$: enumerates all partitions of n ; finds the cyclic ones by iterating B ; checks that they equal the predicted set $\{\lambda(\varepsilon) : \varepsilon \in \{0, 1\}^k, |\varepsilon| = r\}$ (k the rank, $r = n - T_{k-1}$); counts the cycles and compares with $\frac{1}{k} \sum_{d \mid \gcd(k, r)} \varphi(d) \binom{k/d}{r/d}$; and for triangular n checks that every orbit reaches δ_k . It checks nothing beyond $n \leq \text{NMAX}$.

Run from the submission folder (the folder containing `code/`):

```
/usr/bin/time -p python3 code/check_c1.py 45
```

Expected output: 46 lines, one line per $n = 1, \dots, 45$ of the form

```
n=45 k=9 r=9 #partitions=89134 #cyclic=1 #cycles=1 OK
```

followed by the final line

ALL OK up to 45

Measured runtime (Python 3.14.0, this submission's copy of the script): real 19.11 s with the project virtual-environment interpreter and real 20.86 s with the system `python3`; both runs exited with status 0 and produced byte-identical output. This is well under the 10-minute limit.

7 Limitations

- What is established: (i), (ii)(a) and (ii)(b) of Cell C1, for all $k \geq 1$ and $1 \leq r \leq k$. Not addressed: the time-to-cycle questions (cells C2 and beyond).
- The count in (ii)(b) is for the reading of “distinct cycles” fixed in convention 0.1: a cycle is the forward orbit of a cyclic partition, and two cycles are distinct if they are distinct as sets.
- The computer check covers only $1 \leq n \leq 45$ and proves nothing beyond that range; the proof does not rely on it.
- Standard results (pigeonhole principle, Chinese remainder theorem, orbit–stabiliser theorem) are used without proof and without bibliographic reference; see the Cited section.
- No literature search is part of the material this document was prepared from; no claim is made about whether these results appear in the literature.
- Editorial change: in the final paragraph of the proof (“Established / not established”), the accepted proof names the script as `out/code/check_c1.py`; in this submission the script is at `code/check_c1.py`, and the path has been changed accordingly. No other change was made to the text of the proof beyond conversion from Markdown to L^AT_EX.
- This document was typeset by the team's scribe from the accepted artefacts. The scribe added no mathematics and did not referee the proof.

A Listing of code/check_c1.py

```
"""Sanity check for B-C1 (stdlib only, exact integer arithmetic).
For every n in [1, NMAX]: enumerate all partitions of n, compute the cyclic ones by
iterating B (functional graph), compare with the predicted set {lambda(eps)}, count
cycles and compare with (1/k) sum_{d | gcd(k,r)} phi(d) C(k/d, r/d).
Also checks for triangular n that every orbit reaches delta_k.
Not load-bearing: the proof does not rest on this computation."""
import sys
from math import comb, gcd

def partitions(n, maxpart=None):
    if maxpart is None:
        maxpart = n
    if n == 0:
        yield ()
        return
    for p in range(min(n, maxpart), 0, -1):
        for rest in partitions(n - p, p):
            yield (p,) + rest
```

```

def B(lam):
    s = len(lam)
    parts = [x - 1 for x in lam if x > 1] + [s]
    return tuple(sorted(parts, reverse=True))

def phi(m):
    return sum(1 for i in range(1, m + 1) if gcd(i, m) == 1)

def rank(n):
    k = 1
    while k * (k + 1) // 2 < n:
        k += 1
    return k

def predicted(k, r):
    out = set()
    for mask in range(1 << k):
        eps = [(mask >> (j)) & 1 for j in range(k)] # eps[0..k-1] = eps_1..eps_k
        if sum(eps) != r:
            continue
        lam = tuple(x for x in (k - (j + 1) + eps[j] for j in range(k)) if x > 0)
        out.add(lam)
    return out

def main(NMAX):
    for n in range(1, NMAX + 1):
        k = rank(n)
        r = n - (k - 1) * k // 2
        parts = list(partitions(n))
        cyc = set()
        for lam in parts:
            seen = {}
            x = lam
            i = 0
            while x not in seen:
                seen[x] = i
                x = B(x)
                i += 1
            # x is the first repeated element: it lies on the cycle
            y = x
            while True:
                cyc.add(y)
                y = B(y)
                if y == x:
                    break
        pred = predicted(k, r)
        assert cyc == pred, (n, k, r)
        # count cycles
        left = set(cyc)
        ncyc = 0
        while left:
            x = left.pop()
            y = B(x)
            while y != x:
                left.discard(y)
                y = B(y)
            ncyc += 1
        g = gcd(k, r)
        num = sum(phi(d) * comb(k // d, r // d) for d in range(1, g + 1) if g % d == 0)

```

```

assert num % k == 0 and num // k == ncyc, (n, k, r, ncyc, num)
if r == k:
    delta = tuple(range(k, 0, -1))
    assert cyc == {delta}
    for lam in parts:
        x = lam
        for _ in range(len(parts) + 1):
            if x == delta:
                break
            x = B(x)
        assert x == delta, (n, lam)
    print(f"n={n} k={k} r={r} #partitions={len(parts)} #cyclic={len(cyc)} #cycles={ncyc} OK")
print("ALL OK up to", NMAX)

if __name__ == "__main__":
    main(int(sys.argv[1]) if len(sys.argv) > 1 else 40)

```